

BSc (Engineering) in CSE, Level 4, Semester I, Quiz (#2) Examination 2024 (Jan-June)
Course Code: CSE 405, Credit: 3.0, Course Title: Computer Graphics and Image Processing

Time: 30 Minutes

Total: Marks:15

1. Define image restoration. How does it differ from image enhancement? 2+2
2. What is noise modeling in image restoration? Mention two common types of noise found in digital images. 2+2
3. What is edge-based segmentation? Name any two edge detection operators. 2+1
4. An image region is represented by the 3×3 neighborhood of gray levels image I. You are asked to apply for the **Laplacian kernel**.
Compute the Laplacian response at the center pixel and explain how the Laplacian-filtered image can be used to **sharpen** the original image. 2+2

20	18	22
17	<u>19</u>	21
16	20	18

Image I

-1	-1	-1
-1	8	-1
-1	-1	-1

Kernel

$$m(x_2 - x_1) = \frac{y_2 - y_1}{m} + x_1$$

Department of Computer Science and Engineering (CSE)
BSc (Engineering) in CSE

Level 4, Semester I, Mid-term Examination 2024 (Jan-Jun)

Course Code: CSE 405, Credit: 3.0, Course Title: Computer Graphics and Image Processing

Time: 50 Minutes (Answer any three (03) of the following question)

Total: Marks:30

[N.B. The figure in the right margin indicates the marks allocated for respective question.

Split answer of any question is not allowed.]

1. a) Define scan conversion. What key considerations should be kept in mind while performing scan conversion? 2
- b) What is transformation? Briefly explain rotation transformation. 3
- c) Using the Bresenham line algorithm, determine the pixel positions for a line from (3,2) to (9,6). Show the decision parameter calculations for each step. 5
2. a) Mention the limitation of DDA line drawing algorithm. 2
- b) State the homogeneous coordinate representation of a 2-D point (x, y). Why is it useful for composite transformations? 3
- c) Use the Cohen-Sutherland algorithm to clip the line from P1(10, 30) to P2(80, 90) against the rectangular window with A (20, 20), B (90, 20), C (90, 70), and D (20, 70). Show region codes and final clipped coordinates. 5

Section-B

(Answer any 03(three) from the following questions)

1. a) What is spatial filtering, and how does it differ from frequency domain filtering? 3
- b) Find the number of bits required to store a 256 X 256 image with 256 gray levels. 2
- c) What is image restoration? Briefly explain the degradation model. 2+3
- d) Consider the following 5x5 grayscale image and the sharpening filter: 5

10	11	10	10	10
<u>10</u>	20	22	20	20
11	20	<u>15</u>	10	20
10	20	10	10	20
10	20	20	<u>20</u>	25

0	-1	0
-1	5	-1
0	-1	0

Compute the filtered value for the underlined pixel. Also, describe the effect of applying this sharpening filter. [Hint: Assume zero-padding for boundary pixels if required.]

2. a) What is Gaussian noise? 2
- b) Briefly explain the fundamental steps in digital image processing. 4
- c) Define image interpolation. Explain any two image interpolation techniques. 2+2
- d) Apply histogram equalization to the 5x5 image below to perform image enhancement. Sketch the histograms of the original image and histogram equalized image. 5

1	1	5	6	6
3	1	1	5	6
3	5	5	3	3
3	1	5	1	3
1	1	1	1	1

1 - 11
3 - 6
5 - 5
6 - 3

3. a) Explain 4-adjacency and 8-adjacency in a binary image. 3
- b) Find the City-block and Euclidean distance between points P (15, 20) and Q (25, 30). 3
- c) What is spatial correlation and spatial convolution in image processing? Explain the difference. 2+2
- d) Perform the 2D spatial convolution of the following 5x5 image f , with the 3x3 kernel w . Use zero padding and produce a 5x5 output. 5

0	0	0	0	0
0	0	0	0	0
0	0	1	0	0
0	0	0	0	0
0	0	0	0	0

f

9	8	7
6	5	4
3	2	1

w

- A. a) Define image segmentation and its importance in digital image processing. 3
- b) What is the difference between Laplacian and Sobel operators? 3
- c) Show how a Laplacian filter enhances image edges. Provide the mask and explanation. 4
- d) Briefly explain the terms: thresholding and region growing. 5

Section-B

(Answer any 03(three) from the following questions)

1. a) What is spatial filtering, and how does it differ from frequency domain filtering?
- b) Find the number of bits required to store a 256 X 256 image with 256 gray levels.
- c) What is image restoration? Briefly explain the degradation model.
- d) Consider the following 5×5 grayscale image and the sharpening filter:

10	11	10	10	10
<u>10</u>	20	22	20	20
11	20	<u>15</u>	10	20
10	20	10	10	20
10	20	20	<u>20</u>	25

0	-1	0
-1	5	-1
0	-1	0

Compute the filtered value for the underlined pixel. Also, describe the effect of applying this sharpening filter. [Hint: Assume zero-padding for boundary pixels if required.]

2. a) What is Gaussian noise?
- b) Briefly explain the fundamental steps in digital image processing.
- c) Define image interpolation. Explain any two image interpolation techniques.
- d) Apply histogram equalization to the 5×5 image below to perform image enhancement. Sketch the histograms of the original image and histogram equalized image.

1	1	5	6	6
3	1	1	5	6
3	5	5	3	3
3	1	5	1	3
1	1	1	1	1

3. a) Explain 4-adjacency and 8-adjacency in a binary image.
- b) Find the City-block and Euclidean distance between points P (15, 20) and Q (25, 30).
- c) What is spatial correlation and spatial convolution in image processing? Explain the difference.
- d) Perform the 2D spatial convolution of the following 5×5 image f , with the 3×3 kernel w . Use zero padding and produce a 5×5 output.

0	0	0	0	0
0	0	0	0	0
0	0	1	0	0
0	0	0	0	0
0	0	0	0	0

f

9	8	7
6	5	4
3	2	1

w

4. a) Define image segmentation and its importance in digital image processing.
- b) What is the difference between Laplacian and Sobel operators?
- c) Show how a Laplacian filter enhances image edges. Provide the mask and explanation.
- d) Briefly explain the terms: thresholding and region growing.

3. a) An 8-bit image of size 512×512 is to be transmitted. Calculate the number of bits required to store the image. 2
- b) Briefly explain the general pipeline of an image processing system. 3
- b) What is histogram equalization? Draw the histogram of the set of gray levels $\{52, 55, 52, 60, 70, 70, 80, 80, 90, 55, 50, 70, 70, 70, 80, 80\}$ and perform one iteration of histogram equalization. 5
4. a) List some common preprocessing techniques used in image processing. 2
- b) How does a smoothing filter affect an image? Give examples of commonly used smoothing filters. 3
- c) Consider the following 6×6 grayscale image and the sharpening filter: 5

10	20	30	40	50
15	<u>25</u>	35	45	55
20	30	<u>40</u>	50	60
25	35	45	<u>55</u>	65
30	40	50	60	70

0	-1	0
-1	5	-1
0	-1	0

$P_1(20, 47)$

$P_2(79, 70)$

Compute the filtered value for the underlined samples. Also, describe the effect of applying this sharpening filter.

50
52, -11
55, 1
60, 1
70, 1111
80, 1111
90, 1